



19AI301/ CS3301	Python Programming	L	T	P	C
		2	0	2	3

PREAMBLE

This course provides learners, the depth knowledge on python programming. The learners will get the hands-on experience for using various programming constructs for, while, and if to solve engineering problems. The introduction of packages like NumPy and pandas provides the learner confidence to handle large complex problems.

PREREQUISITE:

Nil

COURSE OUTCOMES

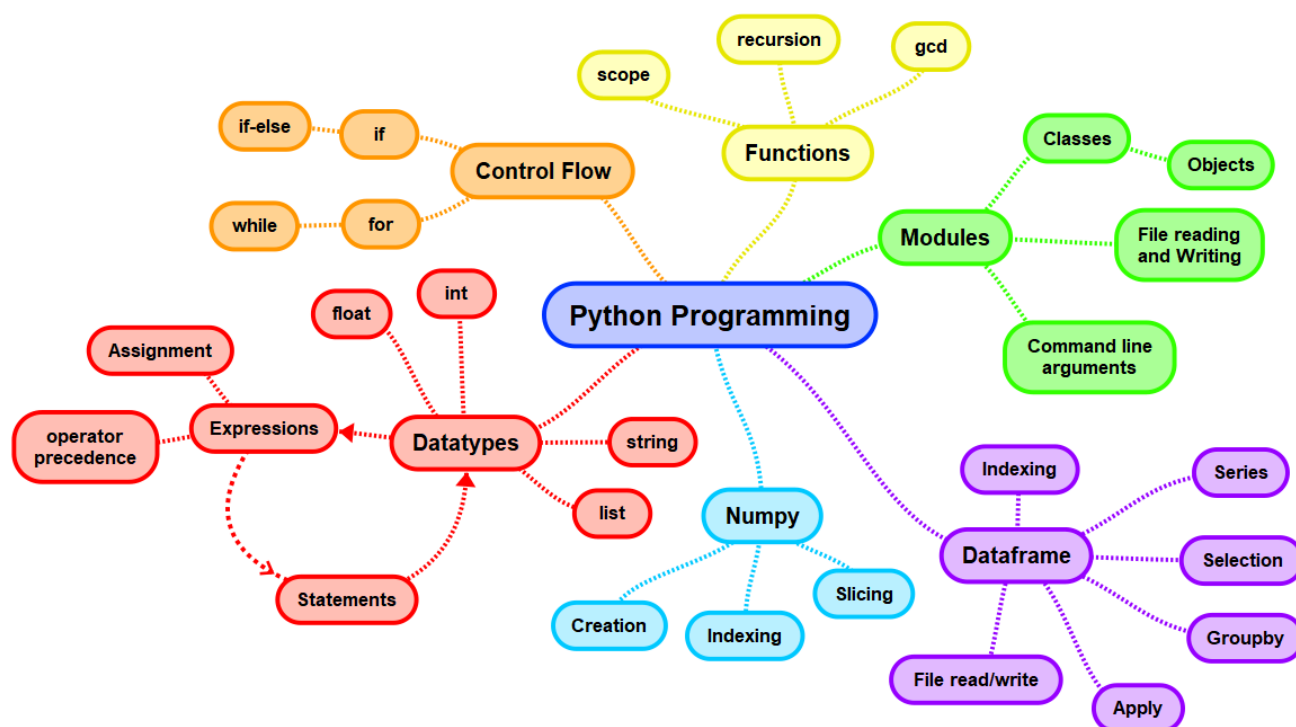
At the end of this course, the learner can able to		
CO1	To read and write simple Python programs.	Understand
CO2	To develop Python programs with conditionals and loops.	Create
CO3	To use Python data structures -- lists, tuples, dictionaries.	Apply
CO4	To use files, modules and packages.	Apply
CO5	To use NumPy and data frame for data processing	Apply

MAPPING OF COs WITH POs & PSOs

COURSE OUTCOMES	PROGRAMME OUTCOMES												PROGRAMME SPECIFIC OUTCOMES			
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O 1	PS O 2	PSO 3	PSO 4
CO1	3	1	1	1	1	-	-	-	1	-	1	-	1	2	-	1
CO2	3	3	3	3	1	-	-	-	1	-	2	-	2	2	-	2
CO3	3	2	2	-	1	-	-	-	1	-	2	1	2	2	-	2
CO4	2	3	3	-	3	1	-	-	2	2	3	3	1	2	-	3
CO5	2	3	3	-	3	1	-	-	2	2	3	3	1	2	-	3
1. LOW 2.MODERATE 3. SUBSTANTIAL																



CONCEPT MAP



SYLLABUS

UNIT-I	DATA TYPES, EXPRESSIONS, STATEMENTS	12
Python interpreter and interactive mode; values and types: int, float, boolean, string, and list; variables, expressions, statements, tuple assignment, precedence of operators, comments; modules and functions, function definition and use, flow of execution, parameters and arguments; Illustrative programs: <i>Exchange the values of two variables, circulate the values of n variables, distance between two points.</i>		
UNIT-II	CONTROL FLOW, FUNCTIONS	12
Conditionals: Boolean values and operators, conditional (if), alternative (if-else), chained conditional (if-elif-else); Iteration: state, while, for, break, continue, pass; Fruitful functions: return values, parameters, local and global scope, function composition, recursion; Strings: string slices, immutability, string functions and methods, string module; Lists as arrays. Illustrative programs: square root, gcd, exponentiation, sum an array of numbers. <i>Compute the GCD of two numbers. Find the square root of a number (Newton's method)</i>		



UNIT-III	LISTS, TUPLES, DICTIONARIES	12
Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters; Tuples: tuple assignment, tuple as return value; Dictionaries: operations and methods; advanced list processing - list comprehension; Illustrative programs: selection sort, insertion sort, merge sort, histogram. <i>Find the maximum of a list of numbers</i> <i>Linear search and Binary search</i> <i>Selection sort, Insertion sort</i>		
UNIT-IV	FILES, MODULES, PACKAGES	12
Files and exception: text files, reading and writing files, format operator; command line arguments, errors and exceptions, handling exceptions, modules, packages; Illustrative programs: word count, copy file. Classes, objects. <i>Programs that take command line arguments (word count)</i>		
UNIT-V	NUMPY, DATA FRAME	12
NumPy - Creating a NumPy Array - The Shape and Reshaping of NumPy Array - Indexing and Slicing of NumPy Array - Maths with NumPy Arrays - Basic arithmetic operations on NumPy array. Pandas Series - Dataframe - Selection and Indexing - Missing data - Merging, Joining, Concatenating - Groupby - Apply Functions - Sorting - File Read and Write Support. <i>Multiply, Inverse matrices with loops/library functions and verify the results.</i> <i>Read and process data from a CSV file.</i>		
		TOTAL: 60 PERIODS

LIST OF EQUIPMENTS NEEDED FOR A BATCH OF THIRTY STUDENTS

SLNO	DESCRIPTION	QTY
1	Hardware – PCs	30
2	Anaconda – Python 3.7 Installation	30



TEXT BOOKS

1	Allen B. Downey, ``Think Python: How to Think Like a Computer Scientist“, 2nd edition, Updated for Python 3, Shroff/O'Reilly Publishers, 2016.
2	Guido van Rossum and Fred L. Drake Jr, —An Introduction to Python – Revised and updated for Python 3.2, Network Theory Ltd., 2011.
3	Jake VanderPlas,” Python Data Science Handbook: Essential Tools for Working with Data”, O'Reilly Media, Inc, 2016.

REFERENCES

1	Wes McKinney, “Python for Data Analysis”, O'Reilly Media, Inc, 2017. Jesús Rogel-Salazar, "Data Science And Analytics With Python", CRC Press, Taylor & Francis Group, 2017.
2	Joel Grus, "Data Science from Scratch", O'Reilly Media, Inc., First Edition, 2015.